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The effect of quantitative and qualitative variables on the price of the car

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✓ Introduction

In this project, our goal is to know how the price of cars is influenced by various factors. We have data on nine explanatory variables: mileage, make, type, cylinder, doors, cruise, sound system quality, and leather interior. Each variable has 804 observations.

To analyze this relationship, we randomly selected a sample of 200 observations. We identified the dependent variable as the "Price of the cars," which is measured in thousands \$.

We categorized our explanatory variables into three types: quantitative variables (such as mileage and number of cylinders), qualitative variables (like make, type, and door count), and dummy variables. Dummy variables take only two values, 0 or 1, and are used to represent qualitative variables in statistical models.

Variables	Type
Price	Quantitative
Mileage	Quantitative
Make	Qualitative
Type	Qualitative
Cylinder	Qualitative
Doors	Quantitative
Cruise	Dummy
Sound	Dummy
Leather	Dummy

✧ Qualitative variables:

The data consists of three qualitative variables: make, type and cylinder:

Make: Manufacturer of the car were registered and classified into Chevrolet, Pontiac, SAAB, Saturn, Buick and Cadillac.

Type: Body type of the car was recorded for each car and it was divided into hatchback, sedan, coupe, wagon and convertible.

Cylinder: It is a part of the engine and is considered an important element for measuring engine's capacity of the car and can be divided into low, medium and high.

When creating a model, you must create (number of categories - 1) dummy variables to avoid the trap of the dummy variable, or create (number of categories) variables while removing the fixed variable from the model, but if preferred, we will use the model that contains (number of categories - 1) variables.

✧ Quantitative variables:

The data consists of two quantitative variables: make, type and cylinder

Mileage: The number of miles that car can travel and it measured in miles.

Doors: The number of doors in the car and it has only two values, either four doors or two doors.

✧ Dummy variables:

Variables arising from qualitative variables, but we measure them so that we can conduct analysis, and they consist of three variables: Cruise, Sound and Leather.

$$\text{Cruise} \begin{cases} 1, & \text{if the car has cruise control} \\ 0, & \text{Otherwise} \end{cases}$$

$$\text{Sound} \begin{cases} 1, & \text{if the car has upgraded sound system} \\ 0, & \text{Otherwise} \end{cases}$$

$$\text{Leather} \begin{cases} 1, & \text{if the car has leather system} \\ 0, & \text{Otherwise} \end{cases}$$

1. Does the mean of the price of the car differ according to the type of the car?

We need to study the relationship between the type of the car and the price, as the type is qualitative and divided into five categories “coupe, hatchback, convertible, wagon, and sedan”, so we will create 5-1 dummy variables”4 dummy variable”:

$D_1 \rightarrow$ Coupe & $D_2 \rightarrow$ Hatchback & $D_3 \rightarrow$ Sedan & $D_4 \rightarrow$ Wagon

➤ Dummy variables:

$D_{1i} \begin{cases} 1, & \text{if the car has coupe type} \\ 0, & \text{Otherwise} \end{cases}$ $D_{2i} \begin{cases} 1, & \text{if the car has hatchback type} \\ 0, & \text{Otherwise} \end{cases}$

$D_{3i} \begin{cases} 1, & \text{if the car has sedan type} \\ 0, & \text{Otherwise} \end{cases}$ $D_{4i} \begin{cases} 1, & \text{if the car has wagon type} \\ 0, & \text{Otherwise} \end{cases}$

The convertible type of car will be the base category.

➤ The equation:

$$\text{Price} = \beta_0 + \beta_1 \text{ Coupe} + \beta_2 \text{ Hatchback} + \beta_3 \text{ Sedan} + \beta_4 \text{ Wagon} + u_i$$

$$y_i = \beta_0 + \beta_1 D_{1i} + \beta_2 D_{2i} + \beta_3 D_{3i} + \beta_4 D_{4i} + u_i$$

➤ The hypotheses tests:

$$H_0: \beta_1 = \beta_2 = \beta_3 = \beta_4 = 0$$

$$H_1: \text{At least one of them} \neq 0$$

➤ R_Output:

```
Call:
lm(formula = data$price ~ data$Type)

Residuals:
    Min       1Q   Median       3Q      Max
-2.8814 -0.8897  0.0961  1.0298  3.3279

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)      4.9188     0.3385  14.531  < 2e-16 ***
data$TypeCoupe      2.7880     0.4022   6.932 5.92e-11 ***
data$TypeHatchback   3.8701     0.4707   8.223 2.75e-14 ***
data$TypeSedan       2.5123     0.3582   7.014 3.71e-11 ***
data$Typewagon       1.6098     0.4413   3.647 0.00034 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.267 on 195 degrees of freedom
Multiple R-squared:  0.297,    Adjusted R-squared:  0.2826
F-statistic: 20.59 on 4 and 195 DF,  p-value: 3.597e-14
```

Output_1 from R “Model_1”

➤ *The estimated model_1:*

$$\widehat{Price} = 4.9188 + 2.7880 \text{ Coupe} + 3.8701 \text{ Hatchback} + 2.5123 \text{ Wagon} + 1.6098 \text{ Sedan}$$

$$\hat{y}_i = 4.9188 + 2.7880 D_{1i} + 3.8701 D_{2i} + 2.5123 D_{3i} + 1.6098 D_{4i}$$

➤ *The interpretations:*

- $\hat{\beta}_0 = 4.9188$ means that the mean average price of convertible car (base category) is 4.9188 thousand \$, since its p-value < 0.01 which means that it is statistically significantly different from zero (significant).
- $\hat{\beta}_1$ (differential intercept) = 2.7880 means that the mean average price of coupe car is more than the mean average price of convertible car by about 2.7880 thousand \$ & since its p-value < 0.01 which means that it is statistically significantly between coupe type and convertible car, then it is statistically significantly different from zero (significant).
- $\hat{\beta}_2$ (differential intercept) = 3.8701 means that the mean average price of hatchback car is greater than the mean average price of convertible car by about 3.8701 thousand \$ & since its p-value < 0.01 then it is statistically significantly between hatchback type and convertible car, which means that it is statistically significantly different from zero (significant).
- $\hat{\beta}_3$ (differential intercept) = 2.5123 means that the mean average price of wagon car is greater than the mean average price of convertible car by about 2.5123 thousand \$ & since its p-value < 0.01 then it is statistically significantly between wagon type and convertible car, which means that it is statistically significantly different from zero (significant).
- $\hat{\beta}_4$ (differential intercept) = 1.6098 means that the mean average price of sedan car is greater than the mean average price of convertible car by about 1.6098 thousand \$ & since its p-value < 0.01 then it is statistically significantly between sedan type and convertible car, which means that it is statistically significantly different from zero (significant).

- $\hat{\beta}_0 + \hat{\beta}_1 = 7.7068$ means the mean average price of coupe car.
- $\hat{\beta}_0 + \hat{\beta}_2 = 8.7889$ means the mean average price of hatchback car.
- $\hat{\beta}_0 + \hat{\beta}_3 = 7.4311$ means the mean average price of wagon car.
- $\hat{\beta}_0 + \hat{\beta}_4 = 6.5286$ means the mean average price of sedan car.

From output_1, we can conclude that since all the coefficients are significant, there is a significant difference between all types of cars (differential intercepts) and the base category which means that the price of the car differs based on the type, so we will reject the null hypotheses (H_0) which means that we have a significant difference between all types and base category at 99% confidence interval.

2. Does the mean of the price of the car differ according to the make of the car, for any car type, and any Cylinder size?

We need to study the relationship and influence of three qualitative variables on the price of the car: make, type and cylinder size.

- Type is qualitative and divided into five categories “Coupe, Hatchback, Convertible, Wagon, and Sedan”, so we will create 5-1 dummy variables “4 dummy variable”:

$D_1 \rightarrow$ Coupe & $D_2 \rightarrow$ Hatchback & $D_3 \rightarrow$ Sedan & $D_4 \rightarrow$ Wagon

The convertible type will be the base category.

- Cylinder is qualitative and divided into three categories “low, medium and weak”, so we will create 3-1 dummy variables “2 dummy variable”:

$C_1 \rightarrow$ low & $C_2 \rightarrow$ moderate

The high cylinder will be the base category.

- Make is qualitative and divided into six categories “Saturn, Pontiac, Cadillac, SAAB, Chevrolet and Buick.” so we will create 6-1 dummy variables “5 dummy variable”:

$M_1 \rightarrow$ Cadillac & $M_2 \rightarrow$ Chevrolet & $M_3 \rightarrow$ Pontiac & $M_4 \rightarrow$ SAAB & $M_5 \rightarrow$ Saturn

The Buick make will be the base category.

➤ *Dummy variables:*

$D_{1i} \begin{cases} 1, & \text{if the car has Coupe type} \\ 0, & \text{Otherwise} \end{cases}$

$D_{2i} \begin{cases} 1, & \text{if the car has hatchback type} \\ 0, & \text{Otherwise} \end{cases}$

$D_{3i} \begin{cases} 1, & \text{if the car has sedan type} \\ 0, & \text{Otherwise} \end{cases}$

$D_{4i} \begin{cases} 1, & \text{if the car has wagon type} \\ 0, & \text{Otherwise} \end{cases}$

$C_{1i} \begin{cases} 1, & \text{if the car has low Cylinder} \\ 0, & \text{Otherwise} \end{cases}$

$C_{2i} \begin{cases} 1, & \text{if the car has moderate Cylinder} \\ 0, & \text{Otherwise} \end{cases}$

$M_{1i} \begin{cases} 1, & \text{if the car has Cadillac make} \\ 0, & \text{Otherwise} \end{cases}$

$M_{2i} \begin{cases} 1, & \text{if the car has Chevrolet make} \\ 0, & \text{Otherwise} \end{cases}$

$M_{3i} \begin{cases} 1, & \text{if the car has Pontiac make} \\ 0, & \text{Otherwise} \end{cases}$

$M_{4i} \begin{cases} 1, & \text{if the car has SAAB make} \\ 0, & \text{Otherwise} \end{cases}$

$M_{5i} \begin{cases} 1, & \text{if the car has Saturn make} \\ 0, & \text{Otherwise} \end{cases}$

➤ *The equation:*

$$\text{Price} = \beta_0 + \beta_1 \text{Coupe} + \beta_2 \text{Hatchback} + \beta_3 \text{Sedan} + \beta_4 \text{Wagon} + \beta_5 \text{Low} + \beta_6 \text{Moderate} \\ + \beta_7 \text{Cadillac} + \beta_8 \text{Chevrolet} + \beta_9 \text{Pontiac} + \beta_{10} \text{SAAB} + \beta_{11} \text{Saturn} + u_i$$

$$y_i = \beta_0 + \beta_1 D_{1i} + \beta_2 D_{2i} + \beta_3 D_{3i} + \beta_4 D_{4i} + \beta_5 C_{1i} + \beta_6 C_{2i} + \beta_7 M_{1i} + \beta_8 M_{2i} + \beta_9 M_{3i} + \beta_{10} M_{4i} + \\ \beta_{11} M_{5i} + u_i$$

➤ *The hypotheses tests:*

$$H_0: \beta_7 = \beta_8 = \beta_9 = \beta_{10} = \beta_{11} = 0$$

$$H_1: \text{At least one of them} \neq 0$$

➤ *R_Output:*

```
Call:
lm(formula = data$price ~ data$Type + data$Cylinder + data$Make)

Residuals:
    Min       1Q   Median       3Q      Max
-1.03340 -0.33774 -0.01759  0.31085  1.78521

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)      5.0179    0.2267  22.139  < 2e-16 ***
data$TypeCoupe    0.5830    0.1830   3.185  0.00169 **
data$TypeHatchback 1.3501    0.2140   6.309  1.97e-09 ***
data$TypeSedan    0.8957    0.1624   5.515  1.14e-07 ***
data$TypeWagon    0.2146    0.1834   1.171  0.24328
data$Cylinderlow  2.5699    0.1480  17.367  < 2e-16 ***
data$Cylindermoderate 0.9717    0.1417   6.857  9.82e-11 ***
data$MakeCadillac -1.0545    0.1926  -5.475  1.39e-07 ***
data$MakeChevrolet  0.4903    0.1525   3.215  0.00154 **
data$MakePontiac   0.3449    0.1530   2.254  0.02537 *
data$MakeSAAB     -2.1456    0.1897 -11.311  < 2e-16 ***
data$MakeSaturn    0.5224    0.1905   2.743  0.00668 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4904 on 188 degrees of freedom
Multiple R-squared:  0.8984,    Adjusted R-squared:  0.8924
F-statistic: 151.1 on 11 and 188 DF,  p-value: < 2.2e-16
```

Output_2 from R "Model_2"

➤ *The estimated model_2:*

$$\widehat{\text{Price}} = 5.0179 + 0.5830 \text{ Coupe} + 1.3501 \text{ Hatchback} + 0.8957 \text{ Sedan} + 0.2146 \text{ Wagon} + \\ 2.5699 \text{ Low} + 0.9717 \text{ Moderate} - 1.0545 \text{ Cadillac} + 0.4903 \text{ Chevrolet} + 0.3449 \text{ Pontiac} - \\ 2.1456 \text{ SAAB} + 0.5224 \text{ Saturn}$$

$$\hat{y}_i = 5.0179 + 0.5830 D_{1i} + 1.3501 D_{2i} + 0.8957 D_{3i} + 0.2146 D_{4i} + 2.5699 C_{1i} + 0.9717 C_{2i} - 1.0545 M_{1i} + 0.4903 M_{2i} + 0.3449 M_{3i} - 2.1456 M_{4i} + 0.5224 M_{5i}$$

➤ *The interpretations:*

- $\hat{\beta}_0 = 5.0179$ means that the mean average price of convertible car (base category) Convertible car, Buick make and high cylinder is about 5.0179 thousand \$ & Since its p-value < 0.01 then it is statistically significantly different from zero (significant).
- $\hat{\beta}_1 = 0.5830$ means that the mean average price of coupe car is greater than the mean average price of convertible car by 0.5830 thousand \$ & Since its p-value < 0.01 then it is statistically significantly between coupe type and convertible type holding the make of car and cylinder size constants, which means that it is statistically significantly different from zero (significant).
- $\hat{\beta}_2 = 1.3501$ means that the mean average price of hatchback car is greater than the mean average price of convertible car by 1.3501 thousand \$ & Since its p-value < 0.01 then it is statistically significantly between hatchback type and convertible type holding the make of car and cylinder size constants, which means that it is statistically significantly different from zero (significant).
- $\hat{\beta}_3 = 0.8957$ means that the mean average price of sedan is greater than the mean average price of convertible car by 0.8957 thousand \$ & Since its p-value < 0.01 then it is statistically significantly between sedan type and convertible type holding the make of car and cylinder size constants, which means that it is statistically significantly different from zero (significant).
- $\hat{\beta}_4 = 0.2146$ as its p-value > 0.01 “insignificant”, so there is no significant difference between the mean average price of wagon and the mean average price of convertible car, holding the make of car and cylinder size constants.
- $\hat{\beta}_5 = 2.5699$ means that the mean average price of the cars with low cylinder is greater than the mean average price of the cars high cylinder by 2.5699 thousand \$ & Since its p-value < 0.01 then it is statistically significantly between low cylinder and low cylinder holding the type of car and make of car constants, which means that it is statistically significantly different from zero (significant).
- $\hat{\beta}_6 = 0.9717$ means that the mean average price of the cars with moderate cylinder is greater than the mean average price of the cars high cylinder by 0.9717 thousand \$, & since its p-value < 0.01 then it is statistically significantly between moderate cylinder and low cylinder holding the type of car and make of car constants, which means that it is statistically significantly different from zero (significant).

- $\hat{\beta}_7 = -1.0545$ means that the mean average price of Cadillac is lower than the mean average price of Buick by 1.0545 thousand \$ & Since its p-value < 0.01 then it is statistically significantly between Cadillac and Buick holding the type of car and cylinder size constants, which means that it is statistically significantly different from zero (significant).
- $\hat{\beta}_8 = 0.4903$ means that the mean average price of Chevrolet is greater than the mean average price of Buick by 0.4903 thousand \$ & Since its p-value < 0.01 then it is statistically significantly between Chevrolet and Buick holding the type of car and cylinder size constants, which means that it is statistically significantly different from zero (significant).
- $\hat{\beta}_9 = 0.3449$ since p-value > 0.01 , it is insignificant, so there is no significant difference between the mean average price of Pontiac and the mean average price of Buick, holding the type of the car and the cylinder size constants.
- $\hat{\beta}_{10} = -2.1456$ means that the mean average price of SAAB is lower than the mean average price of Buick by 2.1456 thousand \$ & Since its p-value < 0.01 then it is statistically significantly between SAAB and Buick holding the type of car and cylinder size constants, which means that it is statistically significantly different from zero (significant).
- $\hat{\beta}_{11} = 0.5224$ means that the mean average price of Saturn cars is greater than the mean average price of Buick cars by 0.5224 thousand \$ & Since its p-value < 0.01 then it is statistically significantly between Saturn and Buick holding the type of car and cylinder size constants, which means that it is statistically significantly different from zero (significant).

From output_2, we can conclude that p-value < 0.01 for all coefficients except for β_4 and β_9 , so we will reject H_0 at $\alpha = 0.01$.

The model is significant, the mean average of the price is differed according to the make of the car, holding the type & the cylinder size constants, except for make Pontiac it has no significant difference between the average price of Pontiac car & the average price of Buick car with 99% C.I, Also, there is no significant difference between the mean average price of wagon and the mean average price of convertible car, holding the make of car and cylinder size constants.

3. Is there an interaction between the type of the type and its cylinder size?

To study whether there is an interaction between the type of the car and the cylinder size, we must estimate the reduced model and the full model, then use F test to test the significant of the interaction parameters.

- Type is qualitative and divided into five categories “Coupe, Hatchback, Convertible, Wagon, and Sedan”, so we will create 5-1 dummy variables “4 dummy variable”:

$D_1 \rightarrow$ Hatchback & $D_2 \rightarrow$ Sedan & $D_3 \rightarrow$ Coupe & $D_4 \rightarrow$ Wagon

The convertible type will be the base category.

- Cylinder is qualitative and divided into three categories “low, medium and weak”, so we will create 3-1 dummy variables “2 dummy variable”:

$C_1 \rightarrow$ low & $C_2 \rightarrow$ moderate

The high cylinder will be the base category.

$D_{1i} \begin{cases} 1, & \text{if the car has hatchback type} \\ 0, & \text{Otherwise} \end{cases}$

$D_{2i} \begin{cases} 1, & \text{if the car has sedan type} \\ 0, & \text{Otherwise} \end{cases}$

$D_{3i} \begin{cases} 1, & \text{if the car has coupe type} \\ 0, & \text{Otherwise} \end{cases}$

$D_{4i} \begin{cases} 1, & \text{if the car has wagon type} \\ 0, & \text{Otherwise} \end{cases}$

$C_{1i} \begin{cases} 1, & \text{if the car has low Cylinder} \\ 0, & \text{Otherwise} \end{cases}$

$C_{2i} \begin{cases} 1, & \text{if the car has moderate Cylinder} \\ 0, & \text{Otherwise} \end{cases}$

➤ The equation reduced Model:

Price= $\beta_0 + \beta_1$ Hatchback + β_2 Sedan + β_3 Coupe + β_4 Wagon + β_5 Low + β_6 Moderate + u_i

$$y_i = \beta_0 + \beta_1 D_{1i} + \beta_2 D_{2i} + \beta_3 D_{3i} + \beta_4 D_{4i} + \beta_5 C_{1i} + \beta_6 C_{2i} + u_i$$

	low	moderate	high
Hatchback	$\beta_0 + \beta_1 + \beta_5$	$\beta_0 + \beta_1 + \beta_6$	$\beta_0 + \beta_1$
Sedan	$\beta_0 + \beta_2 + \beta_5$	$\beta_0 + \beta_2 + \beta_6$	$\beta_0 + \beta_2$
Coupe	$\beta_0 + \beta_3 + \beta_5$	$\beta_0 + \beta_3 + \beta_6$	$\beta_0 + \beta_3$
Wagon	$\beta_0 + \beta_4 + \beta_5$	$\beta_0 + \beta_4 + \beta_6$	$\beta_0 + \beta_4$
convertible	$\beta_0 + \beta_5$	$\beta_0 + \beta_6$	β_0

➤ *R_Output:*

```
Call:
lm(formula = price ~ Hatchback + Sedan + Coupe + Wagon + low +
    moderate, data = df_reg)

Residuals:
    Min       1Q   Median       3Q      Max
-2.9083 -0.4691  0.1255  0.4994  2.3611

Coefficients: (1 not defined because of singularities)
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   3.6081     0.2237   16.126 < 2e-16 ***
Hatchback     2.9925     0.2833   10.565 < 2e-16 ***
Sedan         2.0484     0.1893   10.822 < 2e-16 ***
Coupe         2.2863     0.2186   10.461 < 2e-16 ***
Wagon         NA         NA      NA      NA
low           2.7413     0.1983   13.823 < 2e-16 ***
moderate      1.3587     0.2177    6.242 2.67e-09 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8917 on 194 degrees of freedom
Multiple R-squared:  0.6533,    Adjusted R-squared:  0.6444
F-statistic: 73.11 on 5 and 194 DF,  p-value: < 2.2e-16
```

Output_3 from R "Reduced model_3"

➤ *The estimated reduced Model "Reduced Model_3":*

$$\widehat{Price} = 3.6081 + 2.9925 \text{ Hatchback} + 2.0484 \text{ Sedan} + 2.2863 \text{ Coupe} + 2.7413 \text{ Low} + 1.3587 \text{ Moderate}$$

$$\hat{y}_i = 3.6081 + 2.9925 D_{1i} + 2.0484 D_{2i} - 2.2863 D_{3i} - 2.7413 C_{1i} - 1.3587 C_{2i}$$

➤ *The interpretations reduced Model:*

- $\hat{\beta}_0 = 3.6081$ the mean price of the car if its Convertible car with high cylinder size (The bases category) is about 3.6081 thousand \$ & since its p-value < 0.01 then it is statistically significantly different from zero (significant).
- $\hat{\beta}_1 = 2.9925$ means that the average price of car if the car has hatchback type is greater than the mean the average price of car if the car is Convertible type by about 2.9925 thousand \$, holding cylinder size constant & Since its p-value < 0.01 then it is statistically significantly between hatchback car and Convertible car, which means that it is statistically significantly different from zero (significant).

- $\hat{\beta}_2 = 2.0484$ means that the average price of car if the car has Sedan type is greater than the mean average price of car if the car is Convertible type by about 2.0484 thousand \$, holding cylinder size constant & Since its p-value ≤ 0.01 then it is statistically significantly between Sedan type and Convertible type, which means that it is statistically significantly different from zero (significant).
- $\hat{\beta}_3 = 2.2863$ means that the average price of car if the car has Coupe type is greater than the mean average price of car if the car is Convertible type by about 2.2863 thousand \$, holding cylinder size constant & Since its p-value < 0.01 then it is statistically significantly between Coupe type and Convertible type, which means that it is statistically significantly different from zero (significant).
- $\hat{\beta}_4 = \text{NA}$, since all observations for the Wagon category have the same value or exhibit no variability, the Wagon variable does not provide any unique information to the model. In the cross table of car types "Wagon have just zeros", this lack of variation indicates that Wagon type may not significantly influence the outcome variable compared to other car types.
- $\hat{\beta}_5 = 2.7413$ means that the average price of car if the car has low cylinder size is greater than the mean average price of car if the car is high cylinder size by about 2.7413 thousand \$, holding type of car is constant & Since its p-value < 0.01 then it is statistically significantly between low cylinder size and high cylinder size, which means that it is statistically significantly different from zero (significant).
- $\hat{\beta}_6 = 1.3587$ means that the average price of car if the car has moderate cylinder size is greater than the mean average price of car if the car is high cylinder size by about 0.50627 thousand \$, holding type of car is constant & Since its p-value < 0.01 then it is statistically significantly between moderate cylinder size and high cylinder size, which means that it is statistically significantly different from zero (significant).

From output_3, There is significant difference between Hatchback, Sedan and Coupe with the base category Convertible car, and also there is significant difference between low cylinder size, and moderate cylinder size with the base category high cylinder size.

➤ *The equation full Model:*

$$\text{Price} = \beta_0 + \beta_1 \text{Hatchback} + \beta_2 \text{Sedan} + \beta_3 \text{Coupe} + \beta_4 \text{Wagon} + \beta_5 \text{low} + \beta_6 \text{moderate} + \beta_7 \text{Hatchback*low} + \beta_8 \text{Sedan*low} + \beta_9 \text{Coupe*low} + \beta_{10} \text{Wagon*low} + \beta_{11} \text{Hatchback*moderate} + \beta_{12} \text{Sedan*moderate} + \beta_{13} \text{Coupe*moderate} + \beta_{14} \text{Wagon*moderate} + u_i.$$

$$y_i = \beta_0 + \beta_1 D_{1i} + \beta_2 D_{2i} + \beta_3 D_{3i} + \beta_4 D_{4i} + \beta_5 C_{1i} + \beta_6 C_{2i} + \beta_7 D_{1i}C_{1i} + \beta_8 D_{2i}C_{1i} + \beta_9 D_{3i}C_{1i} + \beta_{10} D_{4i}C_{1i} + \beta_{11} D_{1i}C_{2i} + \beta_{12} D_{2i}C_{2i} + \beta_{13} D_{3i}C_{2i} + \beta_{14} D_{4i}C_{2i} + u_i$$

	low	moderate	high
Hatchback	$\beta_0 + \beta_1 + \beta_5 + \beta_7$	$\beta_0 + \beta_1 + \beta_6 + \beta_{11}$	$\beta_0 + \beta_1$
Sedan	$\beta_0 + \beta_2 + \beta_5 + \beta_8$	$\beta_0 + \beta_2 + \beta_6 + \beta_{12}$	$\beta_0 + \beta_2$
Coupe	$\beta_0 + \beta_3 + \beta_5 + \beta_9$	$\beta_0 + \beta_3 + \beta_6 + \beta_{13}$	$\beta_0 + \beta_3$
Wagon	$\beta_0 + \beta_4 + \beta_5 + \beta_{10}$	$\beta_0 + \beta_4 + \beta_6 + \beta_{14}$	$\beta_0 + \beta_4$
convertible	$\beta_0 + \beta_5$	$\beta_0 + \beta_6$	β_0

➤ R_Output:

```
Call:
lm(formula = price ~ Hatchback + Sedan + Coupe + Wagon + low +
    moderate + Hatchback * low + Sedan * low + Coupe * low +
    Wagon * low + Hatchback * moderate + Sedan * moderate + Coupe *
    moderate + Wagon * moderate, data = df_reg)

Residuals:
    Min       1Q   Median       3Q      Max
-2.84717 -0.39067  0.05728  0.41593  2.42219

Coefficients: (5 not defined because of singularities)
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    4.08898    0.36112   11.323  < 2e-16 ***
Hatchback      1.84755    0.71010    2.602  0.010004 *
Sedan          1.50678    0.44893    3.356  0.000954 ***
Coupe          1.55911    0.46620    3.344  0.000994 ***
Wagon          NA         NA         NA      NA
low            2.15742    0.39793    5.422  1.78e-07 ***
moderate       1.51659    0.49338    3.074  0.002424 **
Hatchback:low  1.58540    0.78684    2.015  0.045325 *
Sedan:low      0.58358    0.49648    1.175  0.241287
Coupe:low      0.96318    0.53330    1.806  0.072490 .
Wagon:low      NA         NA         NA      NA
Hatchback:moderate NA         NA         NA      NA
Sedan:moderate -0.03919    0.57236   -0.068  0.945483
Coupe:moderate NA         NA         NA      NA
Wagon:moderate NA         NA         NA      NA
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8846 on 190 degrees of freedom
Multiple R-squared:  0.6659,    Adjusted R-squared:  0.6501
F-statistic: 42.07 on 9 and 190 DF,  p-value: < 2.2e-16
```

Output_4 from R "Full model_3"

➤ The estimated full Model "full Model_3":

$$\widehat{Price} = 4.08898 + 1.84755 \text{ Hatchback} + 1.50678 \text{ Sedan} + 1.55911 \text{ Coupe} + 2.15742 \text{ low} + 1.51659 \text{ moderate} + 1.58540 \text{ Hatchback*low} + 0.58358 \text{ Sedan*low} + 0.96318 \text{ Coupe*low} - 0.03919 \text{ Sedan*moderate}$$

$$\hat{y}_i = 4.08898 + 1.84755 D_{1i} + 1.50678 D_{2i} + 1.5591 D_{3i} + 2.15742 C_{1i} + 1.51659 C_{2i} + 1.58540 D_{1i}C_{1i} + 0.58358 D_{2i}C_{1i} + 0.96318 D_{3i}C_{1i} - 0.03919 D_{2i}C_{2i}$$

➤ *The interpretations full Model:*

- $\hat{\beta}_0 = 4.08898$ the mean price of the car if *its* Convertible car with high cylinder size (The bases category) is about 4.08898 thousand \$ & since its p-value < 0.01 then it is statistically significantly different from zero (significant).
- $\hat{\beta}_1 = 1.84755$ the mean price of car if the car has hatchback type is higher than the mean price of car if the car has convertible car by about 1.84755 thousand \$, but who are high cylinder & since p-value < 0.01 “significant”, so there is significant difference between the mean average price of hatchback type and the mean average price of convertible car.
- $\hat{\beta}_2 = 1.50678$ means that the average price of car if the car has Sedan type is greater than the mean average price of car if the car *is* Convertible type by about 1.50678 thousand \$, but who are high cylinder size & Since its p-value < 0.01 then it is statistically significantly between Sedan type and Convertible type, which means that it is statistically significantly different from zero (significant).
- $\hat{\beta}_3 = 1.55911$ means that the average price of car if the car has Coupe type is greater than the mean average price of car if the car *is* Convertible type by about 1.55911 thousand \$, but who are high cylinder & Since its p-value < 0.01 then it is statistically significantly between Coupe type and Convertible type, which means that it is statistically significantly different from zero (significant).
- $\hat{\beta}_5 = 2.15742$ means that the average price of car if the car has low cylinder size is greater than the mean average price of car if the car *is* high cylinder size by about 2.15742 thousand \$, but who are convertible type & Since its p-value < 0.01 then it is statistically significantly between low cylinder size and high cylinder size, which means that it is statistically significantly different from zero (significant).
- $\hat{\beta}_6 = 1.51659$ means that the average price of car if the car has moderate cylinder size is greater than the mean average price of car if the car *is* high cylinder size by about 1.51659 thousand \$, but who are convertible type & Since its p-value < 0.01 then it is statistically significantly between moderate cylinder size and high cylinder size, which means that it is statistically significantly different from zero (significant).
- $\hat{\beta}_7 = 1.58540$ the mean price of the car if the car has a hatchback type with low cylinder size is greater than the car with a convertible type and high cylinder size by about 1.58540.
- $\hat{\beta}_8 = 0.58358$ its p-value is > 0.01 “insignificant”, being low cylinder size or moderate cylinder size or high cylinder size doesn’t have effect on the average price of car among whose types of car hatchback and convertible

- $\hat{\beta}_9 = 0.96318$ its p-value is > 0.01 “insignificant”, being low cylinder size or moderate cylinder size or high cylinder size doesn’t have effect on the average price of car among whose types of car Coupe and convertible
- $\hat{\beta}_{12} = -0.03919$ its p-value is > 0.01 “insignificant”, being low cylinder size or moderate cylinder size or high cylinder size doesn’t have effect on the average price of car among whose types of car Sedan and convertible.

> df_reg															
	Hatchback	Sedan	Coupe	Wagon	low	moderate	price								
1	0	0	1	0	0	0	6.258929	105	0	1	0	0	1	0	8.164696
2	0	1	0	0	0	0	7.312485	106	0	0	0	0	1	0	7.597313
3	0	0	0	0	1	0	5.731259	107	0	1	0	0	0	1	7.966636
4	0	1	0	0	0	1	6.631855	108	1	0	0	0	0	1	7.264153
5	0	1	0	0	1	0	10.758953	109	0	1	0	0	0	1	7.436353
6	0	0	1	0	0	0	5.553605	110	0	0	1	0	0	0	5.064307
7	0	1	0	0	0	0	5.252608	111	0	1	0	0	1	0	8.385230
8	0	1	0	0	0	1	8.043314	112	0	1	0	0	0	0	6.706795
9	0	0	1	0	1	0	8.971086	113	0	0	0	0	1	0	5.894813
10	1	0	0	0	1	0	10.036100	114	0	1	0	0	1	0	8.529626
11	0	1	0	0	0	1	7.019562	115	0	1	0	0	0	0	4.549671
12	0	0	1	0	1	0	8.805216	116	0	0	0	0	1	0	5.245840
13	0	0	1	0	1	0	8.090586	117	0	1	0	0	0	1	7.802702
14	0	0	0	0	0	0	4.239455	118	0	0	0	0	1	0	5.603379
15	0	0	1	0	0	0	5.057723	119	0	0	1	0	0	1	8.007741
16	0	0	0	0	1	0	6.090879	120	0	1	0	0	0	1	7.234775
17	0	1	0	0	0	1	6.748698	121	0	1	0	0	0	1	7.417798
18	0	1	0	0	1	0	8.593894	122	0	1	0	0	0	0	5.663045
19	0	0	1	0	0	1	6.784534	123	0	0	0	0	1	0	5.460281
20	0	1	0	0	0	1	6.544478	124	0	0	1	0	0	0	6.038374
21	0	1	0	0	1	0	9.316333	125	0	1	0	0	0	1	7.328245
22	0	1	0	0	0	1	7.780993	126	0	0	0	0	1	0	5.475137
23	0	1	0	0	0	0	5.546193	127	0	0	1	0	0	1	7.062606
24	0	1	0	0	0	1	8.151922	128	0	1	0	0	1	0	8.712444
25	0	0	1	0	0	0	5.325373	129	0	1	0	0	1	0	8.547026
26	1	0	0	0	0	1	7.389839	130	0	0	1	0	1	0	9.447785
27	0	1	0	0	0	1	7.084513	131	0	1	0	0	0	1	6.608989
28	0	1	0	0	0	1	7.600025	132	0	0	1	0	1	0	9.165792
29	0	1	0	0	1	0	9.032438	133	0	1	0	0	0	1	5.109931
30	0	1	0	0	0	1	5.185254	134	0	0	1	0	1	0	8.897848
31	0	0	0	0	1	0	5.298342	135	1	0	0	0	1	0	9.579548
32	0	0	0	0	1	0	5.824049	136	0	1	0	0	0	1	6.194553
33	0	1	0	0	0	1	6.261235	137	0	1	0	0	0	1	7.421044
34	0	1	0	0	0	1	7.218384	138	0	1	0	0	0	0	5.100057
35	0	1	0	0	1	0	9.737332	139	0	1	0	0	0	1	5.971865
36	0	1	0	0	0	1	7.015361	140	0	1	0	0	1	0	6.125267
37	0	0	0	0	1	0	5.718933	141	0	1	0	0	1	0	5.489591
38	0	1	0	0	0	1	6.788753	142	0	0	0	0	1	0	5.629502

Output_ 5 from R “Cross table”

Based on the cross matrix, it appears that the low and moderate categories for Hatchback, Coupe, and Wagon car types have **no variations** in the price column. Specifically:

- For Hatchback, there are no variations in the low and moderate categories.
- For Coupe, there are no variations in the moderate category.
- For Wagon, there are no variations in the low and moderate categories.
- For Wagon, there are just zeros.

This is why the model is returning **NA values** for the interaction terms Hatchback: moderate, Coupe: moderate, and Wagon: moderate, & Wagon. Since there are no variations in the price column for these combinations of car type and price category, the model cannot estimate a coefficient for these interactions.

- $\hat{\beta}_4 = \text{NA}$, since all observations for the Wagon category have the same value or exhibit no variability, the Wagon variable does not provide any unique information to the model. In the cross table of car types "Wagon have just zeros", this lack of variation indicates that Wagon type may not significantly influence the outcome variable compared to other car types.
- $\hat{\beta}_{10} = \text{NA}$, since for Wagon, there are no variations in the moderate category, so, the interaction does not provide any unique information to the model, therefore has no coefficient.
- $\hat{\beta}_{11} = \text{NA}$, since for Hatchback, there are no variations in the moderate category, so, the interaction does not provide any unique information to the model, therefore has no coefficient.
- $\hat{\beta}_{13} = \text{NA}$, since for Coupe, there are no variations in the moderate category, so, the interaction does not provide any unique information to the model, therefore has no coefficient.
- $\hat{\beta}_{14} = \text{NA}$, since for Wagon, there are no variations in the moderate category, so, the interaction does not provide any unique information to the model, therefore has no coefficient.

➤ *The hypotheses tests:*

$$H_0: \beta_7 = \beta_8 = \beta_9 = \dots\dots\dots = \beta_{14} = 0$$

$$H_1: \text{At least one of them} \neq 0$$

➤ *Using F Test:*

$$F_{cal} = \frac{(ESS_{red} - ESS_{full})/P}{ESS_{full}/(n - P)} = \frac{(0.8917 - 0.8846)/8}{0.8846/(200 - 14)} = 0.1867 < F_{0.01,8,186} = 2.51$$

We don't reject H_0 , so β_7 & β_8 & β_{14} are not statistically different from zero "not significant", so there isn't interaction and we should use the reduced model.

4. On average does the make of the car affect the price of the car holding the number of miles the car has made and number of doors constant, and what is the effect of increasing the number of miles the car makes and the number of doors on the price of the car?

Since we have quantitative & qualitative variables, we will use the ANCOVA model, make is qualitative and divided into six categories “Saturn, Pontiac, Cadillac, SAAB, Chevrolet and Buick.” so we will create 6-1 dummy variables “5 dummy variable”:

$M_1 \rightarrow$ Cadillac & $M_2 \rightarrow$ Chevrolet & $M_3 \rightarrow$ Pontiac & $M_4 \rightarrow$ SAAB & $M_5 \rightarrow$ Saturn

$M_{1i} \begin{cases} 1, & \text{if the car has Cadillac make} \\ 0, & \text{Otherwise} \end{cases}$ $M_{2i} \begin{cases} 1, & \text{if the car has Chevrolet make} \\ 0, & \text{Otherwise} \end{cases}$

$M_{3i} \begin{cases} 1, & \text{if the car has Pontiac make} \\ 0, & \text{Otherwise} \end{cases}$ $M_{4i} \begin{cases} 1, & \text{if the car has SAAB make} \\ 0, & \text{Otherwise} \end{cases}$

$M_{5i} \begin{cases} 1, & \text{if the car has Saturn make} \\ 0, & \text{Otherwise} \end{cases}$

The Buick car will be the base category

$X_1 \rightarrow$ the number of miles that car has made & $X_2 \rightarrow$ the number of doors in the car.

➤ **The desired model:**

$$\text{Price} = \beta_0 + \beta_1 \text{ Cadillac} + \beta_2 \text{ Chevrolet} + \beta_3 \text{ Pontiac} + \beta_4 \text{ SAAB} + \beta_5 \text{ Saturn} + \beta_6 \text{ Mileage} + \beta_7 \text{ Doors} + u_i$$

$$y_i = \beta_0 + \beta_1 M_{1i} + \beta_2 M_{2i} + \beta_3 M_{3i} + \beta_4 M_{4i} + \beta_5 M_{5i} + \beta_6 X_{1i} + \beta_7 X_{2i} + u_i$$

➤ **The hypotheses tests:**

To study if on average does the make of the car affect the price of the car holding the number of miles the car has made and number of doors constant, we have to test:

$$H_0: \beta_1 = \beta_2 = \beta_3 = \beta_4 = \beta_5 = 0$$

$$H_1: \text{At least one of them} \neq 0$$

To study what is the effect of increasing the number of miles the car makes on the price of the car holding the number of doors constant, we have to test:

$$H_0: \beta_6 = 0$$

$$H_1: \beta_6 \neq 0$$

To study what is the effect of increasing the number of doors on the price of the car holding the number of miles the car has been driven, we have to test:

$$H_0: \beta_7 = 0$$

$$H_1: \beta_7 \neq 0$$

➤ *R_Output:*

```
Call:
lm(formula = data$price ~ data$Make + data$Mileage + data$Doors)

Residuals:
    Min       1Q   Median       3Q      Max
-2.89187 -0.43517 -0.00863  0.41812  2.22137

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    5.064e+00  3.914e-01  12.936 < 2e-16 ***
data$MakeCadillac -1.736e+00  2.920e-01  -5.946 1.28e-08 ***
data$MakeChevrolet  1.621e+00  2.367e-01   6.847 9.90e-11 ***
data$MakePontiac    5.478e-01  2.536e-01   2.160 0.032043 *
data$MakeSAAB      -9.751e-01  2.671e-01  -3.651 0.000337 ***
data$MakeSaturn     1.734e+00  3.089e-01   5.615 6.83e-08 ***
data$Mileage       2.704e-05  7.566e-06   3.574 0.000444 ***
data$Doors         3.247e-01  7.165e-02   4.532 1.03e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8522 on 192 degrees of freedom
Multiple R-squared:  0.6866,    Adjusted R-squared:  0.6752
F-statistic: 60.09 on 7 and 192 DF,  p-value: < 2.2e-16
```

Output_6 from R “Model_4”

➤ *The estimated model_4:*

$$\widehat{Price} = 5.064 - 1.736 \text{ Cadillac} + 1.621 \text{ Chevrolet} + 0.5478 \text{ Pontiac} - 0.9751 \text{ SAAB} + 1.734 \text{ Saturn} + 0.00002704 \text{ Mileage} + 0.3247 \text{ Doors}$$

$$\hat{y}_i = 5.064 - 1.736 M_{1i} + 1.621 M_{2i} + 0.5478 M_{3i} - 0.9751 M_{4i} + 1.734 M_{5i} + 0.00002704 X_{1i} + 0.3247 X_{2i}$$

➤ *The interpretations:*

- $\hat{\beta}_0 = 5.064$ means that the average price of the Buick car “base category” by about 5.064 thousand \$, holding the number of miles and number of doors are constant & Since its p-value < 0.01, it is statistically significantly different from zero (significant).

- $\hat{\beta}_1 = -1.736$ means that the mean average price of the Cadillac car is lower than the mean average price of the Buick car by about 1.736 thousand \$, holding the number of miles and number of doors are constant & Since its p-value < 0.01 then it is statistically significantly between Cadillac car and Buick car, which means that it is statistically significantly different from zero (significant).
- $\hat{\beta}_2 = 1.621$ means that the mean average price of the Chevrolet car is higher than the mean average price of the Buick car by about 1.621 thousand \$, holding the number of miles and number of doors are constant & Since its p-value < 0.01 then it is statistically significantly between Chevrolet car and Buick car, which means that it is statistically significantly different from zero (significant).
- $\hat{\beta}_3 = 0.5478$ as its p-value > 0.01 “insignificant”, so there is no significant difference between the mean average price of Pontiac car and the mean average price of Buick car, holding the number of miles and number of doors are constant.
- $\hat{\beta}_4 = -0.9751$ means that the mean average price of the SAAB car is lower than the mean average price of the Buick car by about 0.9751 thousand \$, holding the number of miles and number of doors are constant & Since its p-value < 0.01 then it is statistically significantly between SAAB car and Buick car, which means that it is statistically significantly different from zero (significant).
- $\hat{\beta}_5 = 1.734$ means that the mean average price of the Saturn car is higher than the mean average price of the Buick car by about 1.734 thousand \$, holding the number of miles and number of doors are constant & Since its p-value < 0.01 then it is statistically significantly between Saturn car and Buick car, which means that it is statistically significantly different from zero (significant).
- $\hat{\beta}_6 = 0.00002704$ means that for every one-mile increase in the number of miles that car can make, the average price increase by 0.00002704 thousand \$, holding the make of the car and doors are constant& Since its p-value < 0.01 , it is statistically significantly different from zero (significant).
- $\hat{\beta}_7 = 0.3247$ means that for every one-unit increase in the number of doors in the car, the average price increase by 0.3247 thousand \$, holding the make of the car and number of doors are constant& Since its p-value < 0.01 , it is statistically significantly different from zero (significant).

From output_6, we conclude that p-value < 0.01 for all coefficients except for β_3 , so we will not reject H_0 which means that not all coefficients are statistically significantly different from zero.

We have evidence that the make of the car affects the price holding the number of miles that car makes & the number of doors that car have.

Also, for every one-unit increase in the number of miles & the number of doors, the average price of the car increase.

5. The effect of the number of miles on the price of the car differ according to the type of the car?

Type is qualitative and divided into five categories “Coupe, Hatchback, Convertible, Wagon, and Sedan”, so we will create 5-1 dummy variables “4 dummy variable”:

$D_1 \rightarrow$ Coupe & $D_2 \rightarrow$ Hatchback & $D_3 \rightarrow$ Sedan & $D_4 \rightarrow$ Wagon

The convertible type will be the base category.

$$D_{1i} \begin{cases} 1, & \text{if the car has coupe type} \\ 0, & \text{Otherwise} \end{cases} \quad D_{2i} \begin{cases} 1, & \text{if the car has hatchback type} \\ 0, & \text{Otherwise} \end{cases}$$

$$D_{3i} \begin{cases} 1, & \text{if the car has sedan type} \\ 0, & \text{Otherwise} \end{cases} \quad D_{4i} \begin{cases} 1, & \text{if the car has wagon type} \\ 0, & \text{Otherwise} \end{cases}$$

$X \rightarrow$ the number of miles that car can make.

➤ The equation reduced model:

$$\text{Price} = \beta_0 + \beta_1 \text{ Coupe} + \beta_2 \text{ Hatchback} + \beta_3 \text{ Sedan} + \beta_4 \text{ Wagon} + \beta_5 \text{ Mileage} + u_i$$

$$y_i = \beta_0 + \beta_1 D_{1i} + \beta_2 D_{2i} + \beta_3 D_{3i} + \beta_4 D_{4i} + \beta_5 X_i + u_i$$

➤ R_Output:

```
Call:
lm(formula = price ~ Coupe + Hatchback + Sedan + Wagon + Mileage,
    data = data)

Residuals:
    Min       1Q   Median       3Q      Max
-2.7219 -0.8563  0.1258  1.0744  3.1699

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  4.306e+00  4.014e-01  10.727  < 2e-16 ***
Coupe        2.779e+00  3.957e-01   7.024 3.56e-11 ***
Hatchback    3.976e+00  4.646e-01   8.558 3.48e-15 ***
Sedan        2.524e+00  3.524e-01   7.163 1.60e-11 ***
Wagon        1.640e+00  4.343e-01   3.777 0.000211 ***
Mileage      3.011e-05  1.101e-05   2.736 0.006796 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.246 on 194 degrees of freedom
Multiple R-squared:  0.3231,    Adjusted R-squared:  0.3057
F-statistic: 18.52 on 5 and 194 DF,  p-value: 5.055e-15
```

Output_7 from R “Reduced Model_5”

➤ *The estimated reduced model “Reduced Model_5”:*

$$\widehat{Price} = 4.306 + 2.779 \text{ Coupe} + 3.976 \text{ Hatchback} + 2.524 \text{ Sedan} + 1.640 \text{ Wagon} + 0.00003011 \text{ Mileage}$$

$$\hat{y}_i = 4.306 + 2.779 D_{1i} + 3.976 D_{2i} + 2.524 D_{3i} + 1.640 D_{4i} + 0.00003011 X_i$$

➤ *The interpretation reduced model:*

- $\hat{\beta}_0 = 4.306$ mean that the mean average price of Convertible “base category” is equal to 4.306 thousand \$, holding the number of miles constant & since its p-value < 0.01, it is statistically significantly different from zero (significant).
- $\hat{\beta}_1 = 2.779$ means that the mean average price of Coupe cars is higher than the mean average price of Convertible car by about 2.779 thousand \$, holding the number of miles constant & since its p-value < 0.01 then it is statistically significantly between Coupe car and Convertible car, which means that it is statistically significantly different from zero (significant).
- $\hat{\beta}_2 = 3.976$ means that the mean average price of Hatchback cars is higher than the Convertible car by about 3.976 thousand \$, holding the number of miles constant & Since its p-value < 0.01 then it is statistically significantly between Hatchback car and Convertible car, which means that it is statistically significantly different from zero (significant).
- $\hat{\beta}_3 = 2.524$ means that the mean average price of Sedan cars is higher than the mean average price of Convertible car by about 2.524 thousand \$, holding the number of miles constant & Since its p-value < 0.01 then it is statistically significantly between Sedan car and Convertible car, which means that it is statistically significantly different from zero (significant).
- $\hat{\beta}_4 = 1.640$ means that the mean average price of Wagon cars is higher than the mean average price of Convertible car by about 1.640 thousand \$, holding the number of miles constant & Since its p-value < 0.01 then it is statistically significantly between Wagon car and Convertible car, which means that it is statistically significantly different from zero (significant).
- $\hat{\beta}_5 = 0.00003011$ when the number of miles increase by one mile the mean average price increase by 0.00003011 thousand \$, holding the type of car is constant & as its p-value < 0.01 “significant”, so there is significant difference between the average price of a car and the number of miles.

➤ *The equation full model:*

$$\begin{aligned} \text{Price} &= \beta_0 + \beta_1 \text{ Coupe} + \beta_2 \text{ Hatchback} + \beta_3 \text{ Sedan} + \beta_4 \text{ Wagon} + \beta_5 \text{ Mileage} + \\ &\beta_6 \text{ Coupe} * \text{Mileage} + \beta_7 \text{ Hatchback} * \text{Mileage} + \beta_8 \text{ Sedan} * \text{Mileage} + \beta_9 \text{ Wagon} * \text{Mileage} + u_i \\ y_i &= \beta_0 + \beta_1 D_{1i} + \beta_2 D_{2i} + \beta_3 D_{3i} + \beta_4 D_{4i} + \beta_5 X_i + \beta_6 D_{1i} X_i + \beta_7 D_{2i} X_i + \beta_8 D_{3i} X_i + \beta_9 D_{4i} X_i + u_i \end{aligned}$$

➤ *R_output:*

```
Call:
lm(formula = price ~ Coupe + Hatchback + Sedan + Wagon + Mileage +
    Coupe * Mileage + Hatchback * Mileage + Sedan * Mileage +
    Wagon * Mileage, data = data)

Residuals:
    Min       1Q   Median       3Q      Max
-2.6969 -0.8965  0.1496  1.0692  3.1812

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   4.077e+00  7.055e-01   5.779 3.03e-08 ***
Coupe         2.347e+00  9.663e-01   2.428  0.01610 *
Hatchback     4.244e+00  1.021e+00   4.158 4.85e-05 ***
Sedan         2.795e+00  7.694e-01   3.633  0.00036 ***
Wagon         3.216e+00  1.150e+00   2.796  0.00570 **
Mileage       4.132e-05  3.054e-05   1.353  0.17764
Coupe:Mileage  2.078e-05  4.297e-05   0.483  0.62930
Hatchback:Mileage -1.356e-05  4.985e-05  -0.272  0.78597
Sedan:Mileage  -1.336e-05  3.370e-05  -0.397  0.69216
Wagon:Mileage  -8.082e-05  5.411e-05  -1.494  0.13690
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.247 on 190 degrees of freedom
Multiple R-squared:  0.3361,    Adjusted R-squared:  0.3046
F-statistic: 10.69 on 9 and 190 DF,  p-value: 2.335e-13
```

Output_8 from R “full Model_5”

➤ *The estimated full model “full model_5”:*

$$\widehat{Price} = 4.077 + 2.347 \text{ Coupe} + 4.244 \text{ Hatchback} + 2.795 \text{ Sedan} + 3.216 \text{ Wagon} + 0.00004132 \text{ Mileage} + 0.00002078 \text{ Coupe} * \text{Mileage} - 0.00001356 \text{ Hatchback} * \text{Mileage} - 0.00001336 \text{ Sedan} * \text{Mileage} - 0.00008082 \text{ Wagon} * \text{Mileage}$$

$$\hat{y}_i = 4.077 + 2.347 D_{1i} + 4.244 D_{2i} + 2.795 D_{3i} + 3.216 D_{4i} + 0.00004132 X_i + 0.00002078 D_{1i} X_i - 0.00001356 D_{2i} X_i - 0.00001336 D_{3i} X_i - 0.00008082 D_{4i} X_i$$

➤ *The interpretations:*

- $\hat{\beta}_0 = 4.077$ mean that the mean average price of Convertible “base category” is equal to 4.077 thousand \$, holding the number of miles constant & since its p-value < 0.01 , it is statistically significantly different from zero (significant).
- $\hat{\beta}_1 = 2.347$ means that the mean average price of Coupe cars is higher than the mean average price of Convertible car by about 2.347 thousand \$, holding the number of miles constant & since its p-value < 0.01 then it is statistically significantly between Coupe car and Convertible car, which means that it is statistically significantly different from zero (significant).
- $\hat{\beta}_2 = 4.244$ means that the mean average price of Hatchback cars is higher than the Convertible car by about 4.244 thousand \$, holding the number of miles constant & Since its p-value < 0.01 then it is statistically significantly between Hatchback car and Convertible car, which means that it is statistically significantly different from zero (significant).
- $\hat{\beta}_3 = 2.795$ means that the mean average price of Sedan cars is higher than the mean average price of Convertible car by about 2.795 thousand \$, holding the number of miles constant & Since its p-value < 0.01 then it is statistically significantly between Sedan car and Convertible car, which means that it is statistically significantly different from zero (significant).
- $\hat{\beta}_4 = 3.216$ means that the mean average price of Wagon cars is higher than the mean average price of Convertible car by about 3.216 thousand \$, holding the number of miles constant & Since its p-value < 0.01 then it is statistically significantly between Wagon car and Convertible car, which means that it is statistically significantly different from zero (significant).
- $\hat{\beta}_5 = 0.00004132$ as its p-value > 0.01 “insignificant”, so there is no significant difference between the average price of a car and the number of miles, holding type car is constant.
- $\hat{\beta}_6 = 0.00002078$ as its p-value > 0.01 “insignificant” which means it is not statistically significant, the effect of number of miles on average price doesn’t differ from coupe to Convertible cars.
- $\hat{\beta}_7 = -0.00001356$ as its p-value > 0.01 “insignificant” which means it is not statistically significant, the effect of number of miles on average price does not differ from Hatchback to Convertible cars.
- $\hat{\beta}_8 = -0.00001336$ as its p-value > 0.01 “insignificant” which means it is not statistically significant, the effect of number of miles on average price does not differ from Sedan to Convertible cars.

- $\hat{\beta}_9 = -0.00008082$ as its p-value > 0.01 “insignificant” which means it is not statistically significant, the effect of number of miles on average price does not differ from Wagon to Convertible cars.

From output_8, We don't reject H_0 , so β_6 & β_7 & β_8 & β_9 are not statistically different from zero “not significant”, the effect of number of miles that car makes on the price does not differ according to the car type.

➤ *Hypotheses test:*

$$H_0: \beta_6 = \beta_7 = \beta_8 = \beta_9 = 0$$

$$H_1: \text{at least one of them} \neq 0$$

➤ *Using F Test*

$$F_{cal} = \frac{(ESS_{red} - ESS_{full}) / P_{test}}{ESS_{full} / (n - P)} = \frac{(1.246 - 1.247) / 4}{1.247 / (200 - 9)} = -0.0383 < F_{0.01, 4, 191} = 3.32$$

We don't reject H_0 , so β_6 & β_7 & β_8 & β_9 are not statistically different from zero “not significant”, so there isn't interaction and we should use the reduced model.

6. What are the effects of each of cruise, sound and leather on the price of car, is there any interaction between them?

To study this, we decide to estimate one model, this model includes the differential intercepts and interactions terms, and we are going to do two tests first one to know the effect of each cruise, sound and leather on the price of car, and the second one to check the existence of the interaction.

$$D_{1i} \begin{cases} 1, & \text{if the car has cruise control} \\ 0, & \text{Otherwise} \end{cases}$$

$$D_{2i} \begin{cases} 1, & \text{if the car has upgraded sound system} \\ 0, & \text{Otherwise} \end{cases}$$

$$D_{3i} \begin{cases} 1, & \text{if the car has leather seats} \\ 0, & \text{Otherwise} \end{cases}$$

Then the base category is the car hasn't cruise control, upgraded sound system and leather seats

➤ The equation reduced model:

$$\text{Price} = \beta_0 + \beta_1 \text{Cruise} + \beta_2 \text{Sound} + \beta_3 \text{Leather} + u_i$$

$$y_i = \beta_0 + \beta_1 D_{1i} + \beta_2 D_{2i} + \beta_3 D_{3i} + u_i$$

➤ R_Output:

```
Call:
lm(formula = price ~ Cruise + Sound + Leather, data = data)

Residuals:
    Min       1Q   Median       3Q      Max
-3.09518 -0.96219 -0.06929  0.78100  3.06481

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   8.8738     0.2707  32.775 < 2e-16 ***
Cruise       -1.8185     0.2193  -8.294 1.74e-14 ***
Sound         0.3467     0.1962   1.767  0.0788 .
Leather      -0.5036     0.2064  -2.440  0.0156 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.273 on 196 degrees of freedom
Multiple R-squared:  0.2867,    Adjusted R-squared:  0.2757
F-statistic: 26.25 on 3 and 196 DF,  p-value: 2.551e-14
```

Output_9 from R "Reduced Model_6"

➤ The estimated reduced model "Reduced Model_5":

$$\widehat{\text{Price}} = 8.8738 - 1.8185 \text{Cruise} + 0.3467 \text{Sound} - 0.5036 \text{Leather}$$

$$\hat{y}_i = 8.8738 - 1.8185 D_{1i} + 0.3467 D_{2i} - 0.5036 D_{3i}$$

➤ The interpretations reduced model:

- $\hat{\beta}_0 = 8.8738$ the average of the price of the cars which haven't cruise control, upgraded sound system and leather seats by about 8.8738 thousand \$ & since its p-value < 0.01, it is statistically significantly different from zero (significant).
- $\hat{\beta}_1 = -1.8185$ the average of the price of the cars which have cruise control is lower than the average of the price of the other cars (which doesn't have cruise control) by about 1.8185 *thousand* \$, holding the sound and the leather seats of the cars constant & since its p-value < 0.01, it is statistically significantly different from zero (significant).
- $\hat{\beta}_2 = 0.3467$ as its P-value > 0.01 "insignificant" which means that there is no statistical Difference between the mean price of a car that has upgraded sound system and a car that doesn't have one, holding the cruise control and the leather seats of the cars constant.
- $\hat{\beta}_3 = -0.5036$ the average of the price of the cars which have leather seats is lower than the average of the price of the other cars (which doesn't have leather control) by about 0.5036 *thousand* \$, holding the sound and the cruise control of the cars constant & since its p-value < 0.01, it is statistically significantly different from zero (significant).

➤ The equation full model:

$$\text{Price} = \beta_0 + \beta_1 \text{Cruise} + \beta_2 \text{Sound} + \beta_3 \text{Leather} + \beta_4 \text{Cruise} * \text{Sound} + \beta_5 \text{Cruise} * \text{Leather} + \beta_6 \text{Sound} * \text{Leather} + \beta_7 \text{Cruise} * \text{Sound} * \text{Leather} + u_i$$

$$y_i = \beta_0 + \beta_1 D_{1i} + \beta_2 D_{2i} + \beta_3 D_{3i} + \beta_4 D_{1i} D_{2i} + \beta_5 D_{1i} D_{3i} + \beta_6 D_{2i} D_{3i} + \beta_7 D_{1i} D_{2i} D_{3i} + u_i$$

➤ R_Output:

```
Call:
lm(formula = price ~ Cruise + Sound + Leather + Cruise * Sound +
    Cruise * Leather + Sound * Leather + Cruise * Sound * Leather,
    data = data)

Residuals:
    Min       1Q   Median       3Q      Max
-3.2600 -0.8398 -0.0643  0.7550  3.4107

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)      9.2630     0.4427  20.926 < 2e-16 ***
Cruise          -1.7744     0.5481  -3.237  0.00142 **
Sound           -0.6370     0.7667  -0.831  0.40706
Leather         -0.1829     0.7667  -0.239  0.81168
Cruise:Sound      0.2721     0.8675   0.314  0.75412
Cruise:Leather   -1.0999     0.8578  -1.282  0.20133
Sound:Leather     0.1415     1.0187   0.139  0.88967
Cruise:Sound:Leather 1.0809     1.1250   0.961  0.33789
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.252 on 192 degrees of freedom
Multiple R-squared:  0.3236,    Adjusted R-squared:  0.2989
F-statistic: 13.12 on 7 and 192 DF,  p-value: 8.972e-14
```

Output_10 from R "full Model_6"

➤ *The estimated full model “full model_6”:*

$$\widehat{Price} = 9.2630 - 1.7744 \text{ Cruise} - 0.6370 \text{ Sound} - 0.1829 \text{ Leather} + 0.2721 \text{ Cruise*Sound} \\ - 1.0999 \text{ Cruise*Leather} + 0.1415 \text{ Sound*Leather} + 1.0809 \text{ Cruise*Sound*Leather}$$

$$\hat{y}_i = 9.2630 - 1.7744 D_{1i} - 0.6370 D_{2i} - 0.1829 D_{3i} + 0.2721 D_{1i}D_{2i} - 1.0999 D_{1i}D_{3i} - \\ 0.1415 D_{2i}D_{3i} + 1.0809 D_{1i}D_{2i}D_{3i}$$

➤ *The interpretations:*

- $\hat{\beta}_0 = 9.2630$ the average of the price of the cars which haven't cruise control, upgraded sound system and leather seats by about 9.2630 thousand \$ & since its p-value < 0.01, it is statistically significantly different from zero (significant).
- $\hat{\beta}_1 = -1.7744$ the average of the price of the cars which have cruise control is lower than the average of the price of the other cars (which doesn't have cruise control) by about 1.7744 *thousand* \$, holding the sound and the leather seats of the cars constant & since its p-value < 0.01, it is statistically significantly different from zero (significant).
- $\hat{\beta}_2 = -0.6370$ as its P-value > 0.01 “insignificant” which means that there is no statistical Difference between the mean price of a car that has upgraded sound system and a car that doesn't have one, holding the Cruise and the leather seats of the cars constant.
- $\hat{\beta}_3 = -0.1829$ as its P-value > 0.01 “insignificant” which means that there is no statistical difference between the mean price of a car that has leather seats and a car that doesn't have, holding the sound and the Cruise of the car's constant.
- $\hat{\beta}_4 = 0.2721$ as its P-value > 0.01 “insignificant” which means the effect of the upgraded sound system on the average of the price is not statistically significant different between the cars which have Cruise system and which doesn't have Cruise system holding the leather seats of cars constant.
- $\hat{\beta}_5 = -1.0999$ as its P-value > 0.01 Which means that it is not statistically significant, as the effect of leather seats on the average price does not varies from the cars which have cruise system and which doesn't have Cruise system.
- $\hat{\beta}_6 = 0.1415$ as its P- value > 0.01 “insignificant” which mean the effect of the leather seats on the average of the price is not statistically significant different between the cars which have upgraded sound system and which doesn't have upgraded sound system, holding the Cruise of the cars constant.
- $\hat{\beta}_7 = 1.0809$ as its P-value > 0.01 Which means that it is not statistically significant, as the effect of leather seats on the average price does not varies from the cars which have upgraded sound system and which does have upgraded sound system and the cars which have cruise system and which does have Cruise system.

From output_10, the hypothesis testing we will say that there is an effect of cruise on the mean price of a car but there is no effect of upgraded sound system \$ leather seats on the mean price of a car.

From output we notice that the is just one significant coefficient “Cruise”, & for the interaction there is no significant coefficients, so there is no existence of interaction between the dummy(indicator) variables and we decided to use (model without interaction) because it's more appropriate.

➤ *Hypotheses test:*

$$H_0: \beta_4 = \beta_5 = \beta_6 = \beta_7 = 0$$

$$H_1: \text{At least one of them} \neq 0$$

➤ *Using F Test:*

$$F_{cal} = \frac{(ESS_{red} - ESS_{full})/P_{test}}{ESS_{full}/(n - P)} = \frac{(1.246 - 1.247)/4}{1.247/(200 - 7)} = 0.809 < F_{0.01,4,193} = 3.32$$

We don't reject H_0 , so β_4 & β_5 & β_6 & β_7 are not statistically different from zero “not significant”, so there isn't interaction and we should use the reduced model.

7. Does the regression price- number of miles' changes once the number of miles achieves 15 thousand miles?

In this case we want to know if the relationship between the price and number of miles is different before and after the number of miles achieves 15000 miles or not and because we separate the relationship between the price and the number of miles at certain value of mileage (x) so, to describe this relationship we decided to use the piecewise regression model Piecewise regression model (meaning we have two separable regression lines connected by certain value of x)

And we use the dummy variable to check it and test the slope of the model.

$$D_{1i} = \begin{cases} 1, & \text{if } x > 15000 \text{ miles} \\ 0, & \text{Otherwise} \end{cases}$$

➤ The desired model_7:

$$\text{Price} = \beta_0 + \beta_1 \text{ Mile} + \beta_2 (\text{Mile} - 15000)D_i + u_i$$

➤ R-Output:

```
Call:
lm(formula = price ~ Mileage + new)

Residuals:
    Min       1Q   Median       3Q      Max
-3.3858 -1.3953  0.1571  1.2714  3.3101

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  6.864e+00  4.531e-01  15.150  <2e-16 ***
Mileage      2.078e-05  3.666e-05   0.567   0.572
new          5.988e-06  5.061e-05   0.118   0.906
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.489 on 197 degrees of freedom
Multiple R-squared:  0.01806,    Adjusted R-squared:  0.008094
F-statistic: 1.812 on 2 and 197 DF,  p-value: 0.166
```

Output_10 from R "Model_7"

➤ The estimated model:

$$\widehat{Price} = 6.864 + 0.00002078 \text{ Mile} + 0.000005988 (\text{Mile} - 15000) D_i$$

➤ *Hypothesis testing:*

$$H_0: \beta_2 = 0$$

$$H_0: \beta_2 \neq 0$$

➤ *Interpretation:*

- $\hat{\beta}_0 = 6.864$ when the number of miles equal 0, the mean average of the price increases by 6.864 \$ for number of miles less than 15000.
- $\hat{\beta}_1 = 0.00002078$ every one unit increases in the number of miles, the mean average of the price increases by 0.00002078 thousand \$ for number of miles less than 15000.
- $\hat{\beta}_2 = 0.000005988$ the marginal price prediction for the number of miles greater than 15000 is higher than the marginal price prediction for the number of miles less than 15000 by 0.000005988 thousand\$.

OR

- $\hat{\beta}_2 = 0.000005988$ for every one-unit increase in the number of miles, the average price of the cars that can drive greater than 15000 miles is higher than the average of the price of the cars that can drive less than 15000 miles by 0.000005988 thousand \$, p-value = 0.906 > 0.01 then we don't reject H_0 , β_2 is not statistically significantly different from zero, and that is mean there is no break in the regression line and the relationship between the price and number of miles is not change once the number of miles achieves 15000 miles.

From output_10, we don't need to use the piecewise regression model and we don't need to regress Y (price) on X (Mile) using dummy variable So we can describe the relationship between the price and number of miles using the linear regression model.

For checking the result, we can derive the two linear regression model and comparing the results of them from the previous estimated model by putting $D_i = 0$ (the model for number of miles less than or equal 15000)

We can test the slope of the model β_2 by T-Test or by comparing by P-Value to test the existence of piecewise regression

➤ *The estimated model for number of miles less than or equal 15000:*

$$\widehat{Price} = 6.864 + 0.000026768 * \text{Mile}$$

➤ *Interpretation:*

- $\hat{\beta}_0 = 6.864$ when the number of miles equal 0, the average of the price increases by 6.864 thousand \$ for number of miles less than 15000.
- $\hat{\beta}_1 = 0.00002078$ for every one unit increases in the number of miles, the average of the price increases by 0.00002078 thousand \$ for number of miles less than 15000.

➤ *The estimated model for number of miles greater than or equal 15000:*

$$\widehat{Price} = 6.77418 + 0.000026768 * \text{Mile}$$

➤ *Interpretation:*

- $\hat{\beta}_0 = 6.77418$ when the number of miles = 0, the average of the price increases by 6.77418 thousand \$ for number of miles less than 15000.
- $\hat{\beta}_1 = 0.000026768$ for every one unit increases in the number of miles, the average of the price increases by 0.000026768 thousand \$ for number of miles less than 15000.

So from the results of the two model, we notice there is no a big change in the slope and intercept of the two model so as we said before, using the simple linear regression model is more appropriate for describing this relationship.

✓ Conclusion

We can conclude that the type of the car has significant effect on the average price of the car as the price of all car types “*Coupe, hatchback, sedan, wagon*” is significantly higher than the convertible car “base category”.

In the comparison between the price of the cars according to the make while holding type and size constant, we found that the mean average of the price is differed according to the make of the car, except for make Pontiac it has no significant difference between the average price of Pontiac car & the average price of Buick car, Also, there is no significant difference between the mean average price of wagon and the mean average price of convertible car, holding the make of car and cylinder size constants.

While testing whether there is an interaction between the make of the car & its cylinder size using the ANCOVA model, we fitted the full & the reduced model, then we applied F test on the reduced one to test the interaction, then, we don't reject H_0 , so β_7 & β_8 & β_{14} are not statistically different from zero “not significant”, so there isn't interaction and we should use the reduced model.

To study if on average does the make of the car affect the price of the car holding the number of miles the car has made and number of doors constant, we found that all coefficients are statistically significantly different from zero except for β_3 , so we do not have evidence that the make of the car affects the price for all cars make holding the number of miles that car makes & the number of doors that car have. Also, for every one-unit increase in the number of miles & the number of doors, the average price of the car increase.

The effect of number of miles that car makes on the price does not differ according to the car type. Since all the coefficients “differential slope” are not statistically different from zero “not significant”,

To study the effect of the cruise, sound and leather on the average price of the car, we did two tests first one to know the effect of each cruise, sound and leather on the price of car, and the second one to check the existence of the interaction. And the results are: there is an effect of cruise on the mean price of a car but there is no effect of upgraded sound system \$ leather seats on the mean price of a car. Also, there is existence of interaction between the dummy(indicator) variables and we decided to use the model 2 (model with interaction) because it's more appropriate.

To know the relationship between the price and number of miles is different before and after the number of miles achieves 15000 miles or not using the piecewise regression model, the results of the two models are: there is no a big change in the slope and intercept of the two model so as we said before, using the simple linear regression model is more appropriate for describing this relationship.

In conclusion, our analysis reveals that several key factors influence car prices. Car type, make, mileage, and certain features such as cruise control significantly impact the average price of a car. While interactions between variables exist, they do not highly alter the price determinants.

R_Script

Reading and sampling the data

```
library(readxl)
Data <-read_excel ("data car project.xls")
sample_data = Data [sample (1: nrow(Data), size = 200, replace= F),]
```

Saving the sampled data as an excel file

```
install.packages('writexl')
library(writexl)

write_xlsx (sample_data, "C:/Users/yi727/OneDrive/Desktop/Econometrics
Project//Sampled_data.xlsx")
View(sample_data)
```

Model_1 (Linear model with a categorical variable)

```
attach(sample_data)
data <-read_excel("Sampled_data.xlsx")
Model1=lm(data$price~data$Type)
summary(Model1)
```

Model_2 (Linear model with 3 categorical variable)

```
Model2=lm (data$price~data$Type + data$Cylinder + data$Make)
summary(Model2)
```

Model_3 (interaction between 2 categorical variables)

```
Hatchback<- ifelse (data$Type=="Hatchback", 1, 0)
Sedan<- ifelse (data$Type=="Sedan", 1, 0)
Coupe<-ifelse (data$Type=="Coupe", 1, 0)
Wagon<-ifelse (data$Type=="Wagon", 1, 0)
low<-ifelse (data$Cylinder=="low", 1, 0)
moderate<-ifelse (data$Cylinder=="moderate", 1, 0)
df_reg<-data. Frame (Hatchback=Hatchback, Sedan=Sedan, Coupe=Coupe,
Wagon=Wagon, low=low, moderate=moderate, price = data$price)
df_reg

reduced_model3<-lm (price~Hatchback+Sedan+Coupe+Wagon+low+moderate, data=df_reg)
summary(reduced_model3)
full_model3<-lm (price~Hatchback+Sedan+Coupe+Wagon+low+moderate
+Hatchback*low+Sedan*low+Coupe*low+Wagon*low
+Hatchback*moderate+Sedan*moderate+Coupe*moderate+Wagon*moderate
, data=df_reg)
summary(full_model3)
```

Model_4 (ANCOVA model)

```
model4=lm(data$price~data$Make+data$Mileage+data$Doors)
summary(model4)
model4
```

Model_5 (interaction between qualitative variable and quantitative variable)

```
reduced_model5<-lm(price~Coupe+Hatchback+Sedan+Wagon+Mileage, data=data)
summary(reduced_model5)

Coupe<-ifelse(data$Type=="Coupe",1,0)
Hatchback<-ifelse(data$Type=="Hatchback",1,0)
Sedan<-ifelse(data$Type=="Sedan",1,0)
Wagon<-ifelse(data$Type=="Wagon",1,0)

Model5=lm (price~Coupe+Hatchback+Sedan+Wagon
           +Mileage+Coupe*Mileage+Hatchback*Mileage
           +Sedan*Mileage+Wagon*Mileage, data=data)
summary (Model5)
```

Model_6 (interaction between 3 dummy variable)

```
reduced _model6=lm(price~ Cruise + Sound + Leather ,data=data)
summary (reduced _model6)

model6=lm (price~ Cruise + Sound + Leather + Cruise*Sound + Cruise*Leather +
           Sound*Leather+Cruise*Sound*Leather, data=data)
summary(model6)
```

Model_7 (Using a cut-off point for a qualitative variable)

```
Data $dummy <- ifelse (Mileage > 15000,
                      c (1), c (0))
data$minus = Mileage - 15000
data$new = data $minus* data $dummy
attach(data)
Model7=lm(price~Mileage+new)
summary (Model7)
Model7
OR:
dummy= rep (0, length(data$Mileage))
dummy
dummy[data$Mileage>15000] =1
dummy
##create (x-xi)di:
data$xdif = data$Mileage-15000
data
data$Mileage_Dummy=dummy
data
data$x = data$xdif*data$Mileage_Dummy
data
write_xlsx (data, "C:/Users/yi727/OneDrive/Desktop/Econometrics Project//Data.xlsx")
#model 7:
model7=lm (price ~ Mileage + x, data=data)
summary(model7)
model7
```

price	Mileage	Make	Type	Cylinder	Doors	Cruise	Sound	Leather
6.258929	36480	Pontiac	Coupe	high	2	1	1	1
7.312485	24992	Pontiac	Sedan	moderate	4	1	0	0
5.731259	15050	SAAB	Wagon	low	4	1	0	1
6.631855	5690	Pontiac	Sedan	moderate	4	1	1	1
10.75895	25216	Chevrolet	Sedan	low	4	0	0	0
5.553605	9185	Pontiac	Coupe	high	2	1	1	1
5.252608	26250	Cadillac	Sedan	high	4	1	1	1
8.043314	29925	Pontiac	Sedan	moderate	4	1	1	0
8.971086	22771	Chevrolet	Coupe	low	2	0	1	1
10.0361	29680	Chevrolet	Hatchback	low	4	0	0	1
7.019562	33892	Chevrolet	Sedan	moderate	4	1	0	0
8.805216	23200	Chevrolet	Coupe	low	2	1	1	1
8.090586	16951	Saturn	Coupe	low	2	0	1	0
4.239455	31805	Cadillac	Convertible	high	2	1	0	1
5.057723	10717	Chevrolet	Coupe	high	2	1	1	1
6.090879	31773	SAAB	Wagon	low	4	1	0	1
6.748698	17787	Buick	Sedan	moderate	4	1	1	0
8.593894	17343	Chevrolet	Sedan	low	4	0	1	1
6.784534	13457	Chevrolet	Coupe	moderate	2	1	1	0
6.544478	24027	Buick	Sedan	moderate	4	1	0	1
9.316333	34998	Saturn	Sedan	low	4	0	1	1
7.780993	20751	Pontiac	Sedan	moderate	4	1	1	1
5.546193	20910	Cadillac	Sedan	high	4	1	0	1
8.151922	22964	Buick	Sedan	moderate	4	1	1	0
5.325373	21350	Chevrolet	Coupe	high	2	1	1	1
7.389839	17441	Chevrolet	Hatchback	moderate	4	0	1	1
7.084513	19800	Buick	Sedan	moderate	4	1	1	1
7.600025	21420	Chevrolet	Sedan	moderate	4	1	1	1
9.032438	21492	Chevrolet	Sedan	low	4	1	1	1
5.185254	19100	Cadillac	Sedan	moderate	4	1	0	1
5.298342	10340	SAAB	Convertible	low	2	1	1	0
5.824049	21822	SAAB	Wagon	low	4	1	1	1
6.261235	17480	Buick	Sedan	moderate	4	1	1	0
7.218384	29187	Buick	Sedan	moderate	4	1	1	1
9.737332	38866	Chevrolet	Sedan	low	4	1	1	1
7.015361	17583	Pontiac	Sedan	moderate	4	1	1	0
5.718933	22298	SAAB	Wagon	low	4	1	1	1
6.788753	25489	Buick	Sedan	moderate	4	1	1	1
7.286433	8211	Chevrolet	Sedan	moderate	4	1	0	1
9.859048	16521	Chevrolet	Hatchback	low	4	1	1	1
6.902954	27096	Buick	Sedan	moderate	4	1	0	0

7.599762	8221	Buick	Sedan	moderate	4	1	1	1
8.543877	25845	Chevrolet	Coupe	low	2	1	1	1
7.339053	24747	Pontiac	Sedan	moderate	4	1	1	1
9.067655	21770	Saturn	Sedan	low	4	0	1	0
8.030822	24239	Pontiac	Wagon	low	4	1	0	0
9.050655	23512	Chevrolet	Coupe	low	2	1	1	1
7.685993	2973	Pontiac	Wagon	low	4	0	0	0
7.813489	4188	Pontiac	Sedan	low	4	1	1	1
8.009031	21307	Pontiac	Wagon	low	4	0	0	0
9.177196	28398	Saturn	Sedan	low	4	0	0	0
5.704335	22479	SAAB	Convertible	low	2	1	0	0
8.86176	21386	Chevrolet	Coupe	low	2	1	1	1
9.812394	22225	Chevrolet	Hatchback	low	4	0	0	0
6.672522	8928	Pontiac	Sedan	moderate	4	1	0	1
7.1663	6608	Chevrolet	Hatchback	moderate	4	1	1	1
8.407851	20512	Chevrolet	Sedan	low	4	1	1	1
7.635017	21567	Chevrolet	Sedan	moderate	4	1	0	1
8.827648	20618	Pontiac	Coupe	low	2	1	1	1
6.582714	23798	Buick	Sedan	moderate	4	1	0	1
5.782358	19830	SAAB	Wagon	low	4	1	1	1
4.89869	20575	Cadillac	Sedan	high	4	1	0	1
9.616566	266	Chevrolet	Hatchback	low	4	1	0	1
8.504818	26034	Saturn	Coupe	low	2	0	0	1
5.486558	18661	Cadillac	Sedan	high	4	1	0	1
9.399722	11156	Chevrolet	Sedan	low	4	0	1	1
7.832493	25697	Chevrolet	Sedan	moderate	4	1	1	1
5.844535	21056	Cadillac	Sedan	moderate	4	1	1	1
6.565684	2295	Pontiac	Sedan	moderate	4	1	1	1
6.057104	26400	SAAB	Sedan	low	4	1	1	0
8.615887	18910	Chevrolet	Sedan	low	4	1	1	1
6.10492	4695	Buick	Sedan	moderate	4	1	1	0
8.132798	25979	Saturn	Sedan	moderate	4	1	1	0
9.302721	13404	Chevrolet	Sedan	low	4	1	1	0
7.486951	25453	Chevrolet	Sedan	moderate	4	1	0	1
8.361034	21128	Chevrolet	Sedan	low	4	0	1	1
9.159885	7278	Chevrolet	Sedan	low	4	0	0	0
5.860639	21960	SAAB	Wagon	low	4	1	0	1
7.714006	19498	Chevrolet	Sedan	moderate	4	1	0	0
10.10718	22986	Chevrolet	Hatchback	low	4	0	1	1
6.687673	8970	Buick	Sedan	moderate	4	1	1	0
6.202118	21433	SAAB	Sedan	low	4	1	1	1
5.534266	16975	SAAB	Wagon	low	4	1	1	1

8.429853	15233	Chevrolet	Sedan	low	4	0	1	1
5.812795	16193	Pontiac	Coupe	high	2	1	0	1
8.617871	14231	Chevrolet	Coupe	low	2	1	1	1
7.633171	20829	Pontiac	Sedan	moderate	4	1	0	0
8.392377	11322	Chevrolet	Sedan	low	4	1	1	1
3.955533	18200	Cadillac	Convertible	high	2	1	1	1
5.392188	24031	SAAB	Convertible	low	2	1	0	1
8.549289	10210	Chevrolet	Coupe	low	2	1	1	1
3.818959	6420	Cadillac	Convertible	high	2	1	1	1
7.048716	18419	Pontiac	Sedan	moderate	4	1	1	1
7.67229	24905	Chevrolet	Hatchback	moderate	4	1	1	1
7.223539	1480	Chevrolet	Sedan	moderate	4	1	1	1
9.040657	30297	Chevrolet	Coupe	low	2	1	1	1
9.031814	24318	Pontiac	Coupe	low	2	1	1	1
5.739777	11273	SAAB	Sedan	low	4	1	1	0
3.881504	12021	Cadillac	Convertible	high	2	1	1	1
7.567283	11393	Chevrolet	Sedan	moderate	4	1	1	1
5.665327	18805	SAAB	Wagon	low	4	1	1	1
8.41647	12878	Saturn	Sedan	low	4	1	0	0
7.973243	21391	Pontiac	Wagon	low	4	0	1	1
3.803251	7892	Cadillac	Convertible	high	2	1	1	1
8.164696	14504	Chevrolet	Sedan	low	4	0	1	1
7.597313	19894	Pontiac	Wagon	low	4	1	0	0
7.966636	29325	Pontiac	Sedan	moderate	4	1	1	0
7.264153	8687	Chevrolet	Hatchback	moderate	4	0	1	1
7.436353	29420	Chevrolet	Sedan	moderate	4	1	1	1
5.064307	9410	Chevrolet	Coupe	high	2	1	1	1
8.38523	8427	Chevrolet	Sedan	low	4	0	1	1
6.706795	21929	Pontiac	Sedan	high	4	1	1	1
5.894813	48991	SAAB	Convertible	low	2	1	1	1
8.529626	23748	Chevrolet	Sedan	low	4	0	1	1
4.549671	788	Cadillac	Sedan	high	4	1	0	1
5.24584	18195	SAAB	Convertible	low	2	1	1	0
7.802702	14242	Saturn	Sedan	moderate	4	1	0	0
5.603379	16956	SAAB	Wagon	low	4	1	0	1
8.007741	22414	Pontiac	Coupe	moderate	2	0	1	1
7.234775	24008	Buick	Sedan	moderate	4	1	0	0
7.417798	5826	Saturn	Sedan	moderate	4	1	1	1
5.663045	26222	Cadillac	Sedan	high	4	1	0	1
5.460281	20925	SAAB	Convertible	low	2	1	0	1
6.038374	23886	Pontiac	Coupe	high	2	1	0	1
7.328245	25516	Pontiac	Sedan	moderate	4	1	0	1

5.475137	17590	SAAB	Convertible	low	2	1	1	1
7.062606	24665	Chevrolet	Coupe	moderate	2	1	1	0
8.712444	13318	Saturn	Sedan	low	4	0	0	1
8.547026	21611	Chevrolet	Sedan	low	4	1	1	1
9.447785	27364	Chevrolet	Coupe	low	2	1	1	1
6.608989	26272	Chevrolet	Sedan	moderate	4	1	1	1
9.165792	25285	Pontiac	Coupe	low	2	0	1	1
5.109931	16754	Cadillac	Sedan	moderate	4	1	0	1
8.897848	22571	Chevrolet	Coupe	low	2	1	1	1
9.579548	6699	Chevrolet	Hatchback	low	4	0	1	1
6.194553	9795	Buick	Sedan	moderate	4	1	1	0
7.421044	28354	Pontiac	Sedan	moderate	4	1	0	0
5.100057	18661	Cadillac	Sedan	high	4	1	0	1
5.971865	27484	Cadillac	Sedan	moderate	4	1	1	1
6.125267	22590	SAAB	Sedan	low	4	1	1	1
5.489591	9795	SAAB	Sedan	low	4	1	1	1
5.629502	20103	SAAB	Wagon	low	4	1	1	1
4.836988	2846	Cadillac	Sedan	moderate	4	1	0	1
7.774635	24583	Chevrolet	Hatchback	moderate	4	0	1	1
6.9641	24067	Chevrolet	Coupe	moderate	2	1	0	0
7.734994	26328	Chevrolet	Sedan	moderate	4	1	1	1
9.323093	33962	Saturn	Sedan	low	4	0	0	0
7.577047	4463	Pontiac	Wagon	low	4	1	0	1
8.668221	20043	Chevrolet	Sedan	low	4	0	1	1
7.040524	21211	Pontiac	Sedan	moderate	4	1	0	1
5.932189	14613	SAAB	Sedan	low	4	1	0	1
10.17759	19183	Chevrolet	Sedan	low	4	0	0	0
8.627103	20530	Chevrolet	Sedan	low	4	0	1	1
5.857806	31655	SAAB	Convertible	low	2	1	1	1
7.782964	27460	Chevrolet	Sedan	moderate	4	1	1	1
7.382356	23852	Buick	Sedan	moderate	4	1	1	0
8.75911	23976	Pontiac	Sedan	low	4	0	1	1
8.45308	17270	Chevrolet	Sedan	low	4	0	1	1
6.291005	34175	SAAB	Sedan	low	4	1	1	0
6.778288	25794	Pontiac	Sedan	high	4	1	0	1
7.726094	18562	Chevrolet	Sedan	moderate	4	1	1	1
8.579461	35662	Saturn	Sedan	moderate	4	1	0	0
6.120121	23055	SAAB	Sedan	low	4	1	1	1
5.059004	31587	Cadillac	Sedan	high	4	1	0	1
7.13436	21132	Pontiac	Sedan	moderate	4	1	1	1
7.126561	27077	Pontiac	Sedan	moderate	4	1	1	0
5.470286	6598	Cadillac	Sedan	moderate	4	1	1	1

8.39895	21627	Chevrolet	Sedan	low	4	0	1	1
6.554808	21616	SAAB	Wagon	low	4	1	1	0
7.346323	30063	Pontiac	Sedan	moderate	4	1	0	1
8.746111	14311	Chevrolet	Sedan	low	4	0	1	1
8.925345	32844	Chevrolet	Coupe	low	2	1	1	1
9.928302	23963	Chevrolet	Sedan	low	4	1	1	0
6.676287	22287	Buick	Sedan	moderate	4	1	1	1
8.776675	20309	Chevrolet	Sedan	low	4	1	1	1
7.451522	15190	Chevrolet	Hatchback	moderate	4	0	1	1
8.547321	18766	Chevrolet	Sedan	low	4	1	1	1
9.067029	12101	Chevrolet	Hatchback	low	4	0	0	1
7.996409	8507	Saturn	Coupe	low	2	1	0	0
7.810611	18096	Pontiac	Wagon	low	4	1	1	0
7.15377	22628	Pontiac	Sedan	moderate	4	1	1	0
5.677341	13678	Pontiac	Coupe	high	2	1	1	1
9.292017	36566	Chevrolet	Sedan	low	4	0	1	1
7.004465	25240	Chevrolet	Coupe	moderate	2	1	1	1
8.823279	22382	Chevrolet	Coupe	low	2	1	1	1
6.512459	19148	Pontiac	Sedan	high	4	1	0	1
4.835206	14546	Chevrolet	Convertible	high	2	1	1	1
10.22551	19273	Chevrolet	Sedan	low	4	1	1	0
6.044425	24960	Pontiac	Coupe	high	2	1	1	1
6.10935	25357	SAAB	Wagon	low	4	1	0	1
8.454248	21020	Saturn	Coupe	low	2	0	1	0
8.767888	7372	Chevrolet	Coupe	low	2	1	1	1
9.521163	20156	Chevrolet	Hatchback	low	4	0	1	1
6.741243	15516	Pontiac	Sedan	moderate	4	1	1	1
5.781759	22105	SAAB	Wagon	low	4	1	0	1
7.917626	26070	Chevrolet	Sedan	moderate	4	1	1	1
8.891397	3629	Chevrolet	Sedan	low	4	0	1	0
9.515136	24568	Chevrolet	Hatchback	low	4	1	0	1
5.945248	22328	SAAB	Sedan	low	4	1	1	0
8.452107	18257	Saturn	Sedan	low	4	0	1	1